

ClearEval: A Data-driven Benchmark for Evaluating Large Language Models in Tissue Optical Clearing

Anonymized Authors

Anonymized Affiliations
email@anonymized.com

Abstract. Tissue Optical Clearing (TOC) enables high-resolution 3D visualization, yet complex protocol design risks irreversible specimen damage. While LLMs excel at knowledge retrieval, their ability to synthesize viable wet-lab protocols remains under-evaluated. We introduce ClearEval, the first expert-curated benchmark for TOC protocol design. Moving beyond semantic similarity, we evaluate 13 SOTA LLMs via a dual-track pipeline: static knowledge is assessed via MCQs, while practical generation is evaluated through our novel Completeness, Correctness, and Effectiveness (CCE) framework. CCE serves as an empirically grounded, text-reference-free metric, assessing viability through formulas derived from experimental confidence intervals rather than static texts. Results show that while models exceed 80% accuracy in static retrieval, performance drops significantly for executable protocols. Challenges in incubation time estimation and chemical incompatibility expose a critical “knowledge-execution gap” in scientific AI. This strategy provides a robust validation framework for future scientific agents. The dataset and code are publicly available at <https://github.com/404NotFound422/ClearEval-public>.

Keywords: Benchmark · Tissue optical clearing · Wet-lab protocols · Large language models

1 Introduction

Tissue Optical Clearing (TOC) homogenizes refractive indices to enable non-destructive, sub-cellular 3D imaging via light-sheet microscopy [14, 2], preserving long-range connectivity essential for high-fidelity organ atlases [16, 12]. However, TOC protocol design is a complex multi-objective optimization problem where parameter misjudgments risk irreversible specimen damage, making traditional trial-and-error approaches both costly and inefficient.

While LLMs demonstrate proficiency in biomedical question answering [13], existing scientific benchmarks (e.g., MedQA [6], MMLU [5]) and their recent extensions [18, 15, 3] predominantly assess static knowledge recall without grounding in experimental validation. Furthermore, traditional semantic metrics like

BLEU [11] and general-purpose evaluators [9, 8] fail to capture functional executability. Linguistically fluent protocols can remain experimentally invalid due to incorrect concentrations or processing times—subtle wet-lab errors that generic evaluators often overlook, thereby propagating hallucinations [1, 19]. Evaluating scientific LLMs thus requires a paradigm shift: from text-based similarity to rigorous protocol validation driven by empirical evidence.

To bridge this gap, we introduce **ClearEval**, a benchmark for wet-lab grounded TOC protocol design that integrates an evaluation pipeline constructed from extensive experimental data. We propose the **CCE (Completeness, Correctness, Effectiveness) framework**, which quantitatively validates protocol executability against empirical confidence intervals. Our main contributions are: (1) a novel benchmark paradigm for evaluating the functional executability of LLM-generated protocols; (2) a curated dataset comprising 2,589 domain-specific MCQs and 253 experiment-driven tasks; and (3) the data-driven CCE framework that leverages formulaic assessment to measure LLM performance gaps in practical planning.

2 Related Works

Scientific LLM evaluation has largely focused on static knowledge retrieval [5, 6, 13] or multidimensional reasoning via frameworks like SeedBench [17]. While recent advances [10, 4] highlight the need to bridge computational predictions and experimental validation, existing metrics like BLEU [11] and G-Eval [9] may not fully capture functional executability. This discrepancy often allows linguistically coherent but experimentally invalid protocols to remain undetected, potentially masking hallucinations [1] regarding chemical compatibility or incubation requirements. Unlike text-centric approaches, ClearEval proposes a text-reference-free paradigm grounded in the empirical parameters of Tissue Optical Clearing [14, 2, 7].

3 The ClearEval Benchmark

The development of ClearEval follows a structured pipeline that transforms specialized TOC domain expertise into a computational evaluation framework (Fig. 1).

3.1 Knowledge Base and Task Formulation

ClearEval formalizes wet-lab constraints into a structured knowledge system by integrating curated metadata on tissue types, molecular markers, and physico-chemical properties from over 30 mainstream TOC articles. To align user requirements with protocol selection, these constraints are mapped into a 6-dimensional vector space $\mathcal{V} \in [0, 1]^6$ representing key performance indicators such as fluorescence preservation, reagent permeability, and biosafety. Within this space, 60 core scientific missions across various biological scales were defined by experts to serve as seeds for realistic task generation.

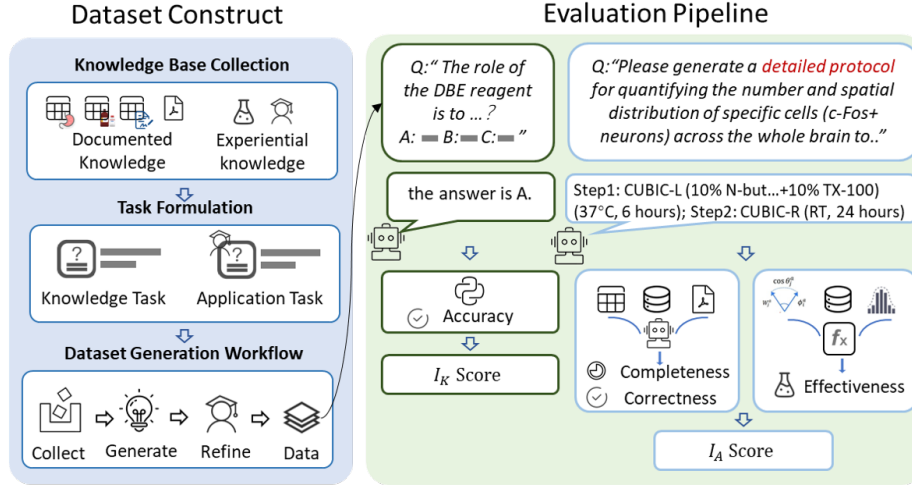


Fig. 1. Overview of the dataset construction and benchmarking pipeline. The left panel shows the workflow for formulating domain knowledge into evaluation datasets. The right panel details the evaluation process, scoring knowledge tasks on accuracy (I_K) and application tasks on completeness, correctness, and a data-driven effectiveness metric (f_x) grounded in experimental confidence intervals to calculate the I_A Score.

3.2 Dataset Generation Workflow

The ClearEval pipeline (Fig. 1, bottom left) utilizes a DeepSeek-driven RAG system to synthesize task candidates from 70 decision templates, 60 mission seeds, and 437 literature-derived data chunks. Each candidate underwent rigorous expert validation to ensure scientific integrity and eliminate hallucinations. The resulting dataset comprises **2,589 Knowledge Tasks (MCQs)** featuring an 8-option format to minimize the probability of correct guessing, and **253 Application Tasks (OEQs)**. Notably, the OEQs move beyond static reference texts, utilizing instead a series of constrained scenarios evaluated through an automated, experiment-based scoring pipeline (Fig. 1, right).

4 Evaluation Framework

To quantify the transition from theoretical recall to functional executability, ClearEval employs a hierarchical scoring system (Table 1). The Application Index (I_A) shifts the focus from semantic similarity to **empirically grounded verification**, utilizing experimental confidence intervals to evaluate the viability of generated protocols.

Table 1. The Hierarchical Metric System of ClearEval. Evaluation scales from static knowledge retrieval to empirically-grounded protocol execution.

Level	Dimension	Sub-Metric	Method	Max
Knowledge (I_K)	Static Retrieval Accuracy		Deterministic	100%
Application (I_A)	Completeness	Step Integrity	LLM-Judge	40%
		Param Detail	LLM-Judge	60%
	Correctness	Sequential Logic	LLM-Judge	40%
		Consistency	LLM-Judge	20%
		Physics Plausibility	LLM-Judge	40%
	Effectiveness	Method Suitability	Vector Sim.	30%
		Label Compatibility	Rule-Based	40%
		Trans. Validity (S_{trans})	Physics Eq.	15%
		Time Efficiency (S_{time})	Physics Eq.	15%
Overall	Expert Score	Harmonic Mean	Aggregation	100%

4.1 Knowledge (I_K) and Application (I_A) Indices

The **Knowledge Index** (I_K) represents the macro-averaged MCQ accuracy across domain-specific categories. To evaluate protocol design, the **Application Index** (I_A) adopts a bottleneck formulation:

$$I_A = \min(\text{Completeness}, \text{Correctness}, \text{Effectiveness}) \quad (1)$$

This "weakest-link" approach reflects wet-lab realities, where a single critical failure in logic or chemical compatibility can render an entire experiment invalid.

4.2 Empirically-Grounded Effectiveness (E_{score})

This dimension quantitatively verifies protocol feasibility against expert databases and physical constraints, yielding $E_{score} = S_{method} + S_{label} + S_{trans} + S_{time}$:

1. Method & Label Suitability: S_{method} evaluates the cosine similarity between the generated and expert constraint vectors:

$$S_{method} = 5 \times \text{ReLU}(\cos(\mathbf{v}_{pred}, \mathbf{v}_{ref})). \quad (2)$$

S_{label} is a **hard constraint**, scoring 0 if any fluorescence quenching risk is detected.

2. Transparency (S_{trans}) & Time Efficiency (S_{time}): To evaluate continuous parameters against **experimental confidence intervals**, we penalize Refractive Index (RI) mismatches and incubation time deviations (Δt) using Gaussian functions:

$$S_{trans} = 3 \cdot \exp\left(-\frac{(RI_{pred} - RI_{ref})^2}{2\sigma_{RI}^2}\right), \quad S_{time} = 3 \cdot \exp\left(-\frac{\Delta t^2}{2\tau^2}\right) \quad (3)$$

where σ_{RI} and τ are expert-defined tolerance parameters. For S_{time} , Δt measures the deviation from the physically viable time window $[t_{min}, t_{max}]$. The raw E_{score} , which aggregates the Empirically-Grounded equations to a theoretical maximum of 17 points, is subsequently min-max normalized to a $[0, 100]$ scale. This aligns it with the Completeness and Correctness dimensions before computing the final I_A bottleneck.

4.3 Overall Expert Score

To penalize models that memorize facts but fail under practical physical constraints, the **ClearEval Expert Score** is the harmonic mean: $\text{Score} = 2I_K I_A / (I_K + I_A)$. Given the bottleneck formulation of I_A , a high score demands both broad theoretical recall and strict experimental compliance.

5 Experiments

5.1 Models and Settings

We benchmarked LLMs across the GPT, Gemini, Claude, DeepSeek, and Qwen families. We set the temperature to 0.0 for **Knowledge tasks** to ensure reproducibility, and used official default temperatures for **Application tasks** to enable reasoning models to fully explore Chain-of-Thought (CoT).

5.2 Main Results: The Knowledge-Execution Gap

Table 2 reveals a stark capability imbalance between static recall and wet-lab execution. This is visually highlighted in Fig. 2: the left panel demonstrates that all evaluated models exhibit uniformly high and tightly clustered proficiency across Knowledge dimensions. In stark contrast, Application Performance (right) is highly erratic. While models generally maintain basic procedural completeness, their trajectories scatter unpredictably and collapse sharply on Effectiveness metrics (e.g., exact RI matching, viable diffusion times), exposing a severe knowledge-execution gap.

5.3 Human Alignment of the Automated Metrics

We validated our automated evaluation pipeline against blinded expert judgments on a subset of 260 generated protocols. As shown in Fig. 3a, alignment varies depending on the underlying evaluation mechanism. Dimensions relying on generic LLM-as-a-judge (Completeness and Correctness) exhibit moderate-to-strong correlations (Pearson $r = 0.77$ and 0.69 , respectively), reflecting the inherent instability of LLMs in assessing complex wet-lab logic. Crucially, our core proposed dimension—the Empirically-Grounded Effectiveness—achieves a strong correlation ($r = 0.83$), demonstrating that mathematical constraints provide a far more reliable proxy for expert judgment.

Table 2. Performance of state-of-the-art models on ClearEval. Evaluation covers Knowledge (MCQ) and Application (OEQ). I_K represents the average of the Knowledge subset, I_A represents the minimum score across the Application subset, and Total is their harmonic mean.

Models	Knowledge (MCQ)				Application (OEQ)				Total
	Tissue	Reagent	Method	I_K	Com.	Cor.	Eff.	I_A	
<i>Proprietary Models</i>									
GPT-5.2-Fast	91.1	88.7	93.5	91.1	<u>87.1</u>	90.2	66.4	66.4	76.8
GPT-5.2-Think	<u>90.5</u>	<u>87.3</u>	<u>93.6</u>	<u>90.5</u>	80.4	88.3	64.1	64.1	75.0
Gemini-3-Flash	78.7	81.8	91.1	83.9	84.6	83.3	66.4	66.4	74.1
Gemini-3-Pro	88.8	77.2	86.7	84.2	81.7	84.8	66.3	66.3	74.2
Claude-4.6-Sonnet	87.6	82.9	94.9	88.5	100.0	85.8	54.2	54.2	67.2
<i>Open-Weight Models</i>									
GLM-4.7	79.3	80.2	93.5	84.3	81.9	<u>89.6</u>	<u>69.0</u>	<u>69.0</u>	75.9
GLM-4.7-Think	78.7	70.5	90.7	80.0	78.8	87.1	70.3	70.3	74.8
DeepSeek-V3.2	79.9	77.9	91.7	83.2	84.6	76.2	62.2	62.2	71.2
DeepSeek-V3.2-Think	85.3	83.8	93.5	87.5	75.0	71.9	62.3	62.3	72.8
Qwen3-max	79.9	<u>87.3</u>	93.1	86.8	<u>87.1</u>	81.0	68.1	68.1	<u>76.3</u>
Qwen3-235B-A22B	77.5	81.2	89.1	82.6	64.6	67.1	53.9	53.9	65.2
Qwen3-32B	73.4	78.6	89.9	80.6	56.2	46.5	55.2	46.5	59.0
Qwen3-14B	73.4	77.8	88.4	79.9	48.8	39.0	54.3	39.0	52.4

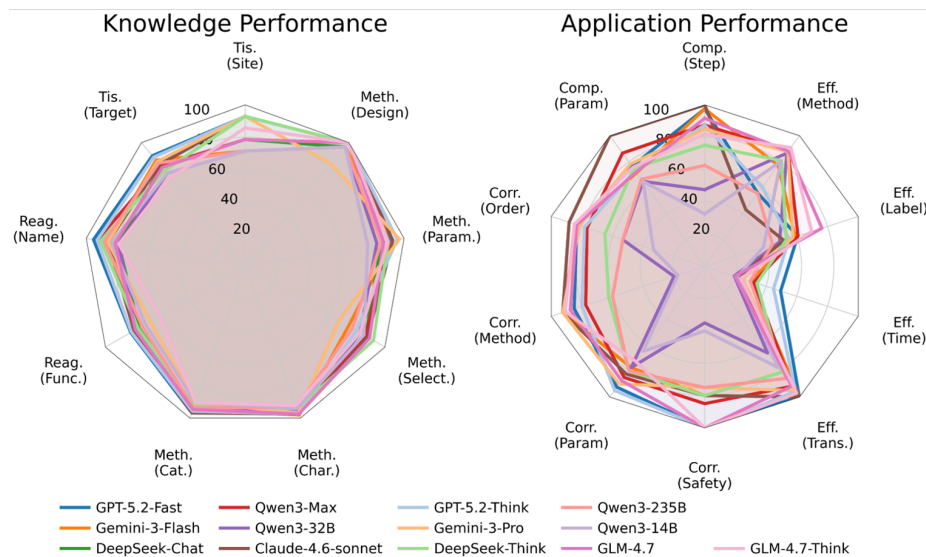


Fig. 2. Performance comparison of 13 LLMs on the ClearEval benchmark. The left panel shows Knowledge Performance (I_K) across domain-specific attributes (Tissue, Reagent, Method). The right panel details Application Performance (I_A), evaluated on completeness, correctness, and an Empirically-Grounded effectiveness metric.

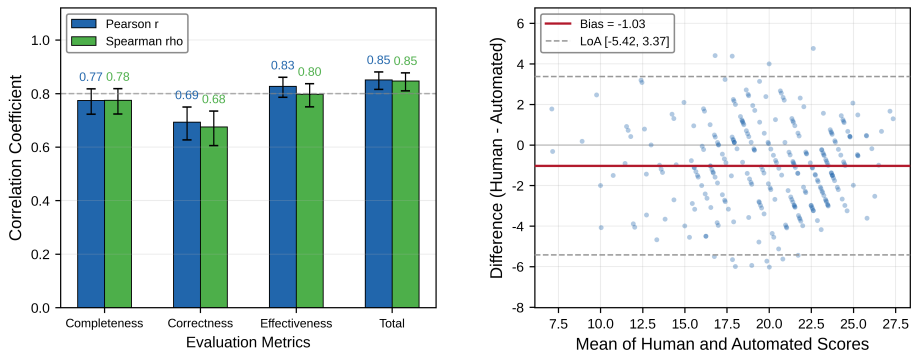


Fig. 3. Human alignment of automated metrics. (a) ClearEval metrics show strong correlation with expert ratings. (b) Bland-Altman plot reveals high absolute agreement with a minimal bias of -1.03.

5.4 Fine-Grained Diagnosis: Effectiveness Bottlenecks

Analysis of specific failure modes (Fig. 4) reveals that the Effectiveness bottleneck is primarily driven by Time Efficiency (S_{time}) and Label Compatibility (S_{label}). In **Time Efficiency** (S_{time}), models exhibit significant *scale-insensitivity*. While ground-truth missions span a 0–200 hour window, generated durations occasionally reach 800 hours. This reflects a systematic bias: LLMs tend to underestimate incubation requirements for large intact tissues while overestimating those for thin sections. Regarding **Label Compatibility** (S_{label}), the broad score distributions indicate inconsistent performance across chemical contexts. While models successfully recall high-frequency knowledge—such as protein quenching by organic solvents—they struggle to manage complex compatibilities between diverse synthetic dyes and specific TOC methodologies.

6 Discussion and Limitations

ClearEval demonstrates that the primary bottleneck in scientific AI is not declarative knowledge retrieval, but the absence of empirical grounding. This knowledge-execution gap suggests that for LLMs to become viable in specialized domains, they must transition into grounded scientific agents that internalize experimental constraints—such as reagent diffusion scales—within their reasoning loops. Such agents could serve as the decision-making core for automated laboratory instruments, ensuring protocols are functionally executable rather than merely linguistically plausible.

However, ClearEval remains a static paradigm. Real-world wet labs involve stochastic variables, such as reagent degradation or tissue heterogeneity, which are challenging to model in silico. Future research should integrate dynamic hardware feedback, allowing agents to iteratively optimize protocols based on

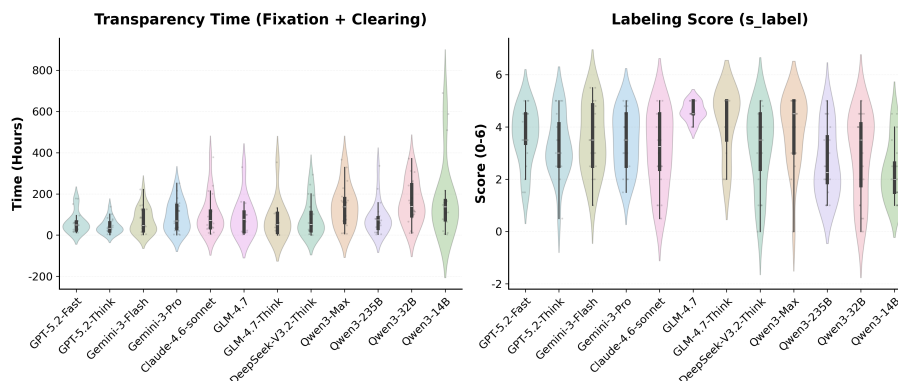


Fig. 4. Performance distributions of LLMs regarding transparency time estimation and labeling compatibility.

real-time experimental signals. This evolution will move LLMs toward becoming experiment-aware collaborators in the broader AI4Science movement.

7 Conclusion

We presented ClearEval, an expert-curated benchmark for TOC protocol design that shifts evaluation from semantic similarity to empirically grounded functional validation. By leveraging a dual-track pipeline and a data-driven CCE framework, ClearEval exposes a significant knowledge-execution gap in current LLMs. Our results indicate that while models excel at knowledge recall, they struggle with the scaling and compatibility constraints required for wet-lab execution. We anticipate that ClearEval will facilitate the development of reliable scientific agents capable of managing complex 3D biomedical imaging workflows.

References

1. Bang, Y., Lee, N., et al.: HalluLens: LLM hallucination benchmark. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (2025)
2. Chung, K., Deisseroth, K.: Structural and molecular interrogation of intact biological systems. *Nature* **497**(7449), 332–337 (2013)
3. Feng, S., Liu, Y., et al.: MediQ: Question-asking LLMs and a benchmark for reliable medical question answering. In: Advances in Neural Information Processing Systems (2024)
4. Gupta, A., Patel, N., et al.: WetBench: LLM-based simulation environment to evaluate wet-lab protocol design. In: International Conference on Learning Representations (2025)
5. Hendrycks, D., Burns, C., Basart, S., et al.: Measuring massive multitask language understanding. In: International Conference on Learning Representations (2021)

6. Jin, D., Pan, E., Oufattole, N., et al.: What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences* **11**(14), 6421 (2021)
7. Kumar, R., Singh, P., et al.: Tissue clearing and its applications in human tissues: A review. *WIREs Mechanisms of Disease* (2024)
8. Li, J., Chen, X., et al.: A survey on LLM-as-a-Judge. *arXiv preprint arXiv:2411.15594* (2024)
9. Liu, Y., Iter, D., Xu, Y., et al.: G-eval: NLG evaluation using gpt-4 with better human alignment. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. pp. 2511–2522 (2023)
10. Martinez, L., Garcia, C., et al.: AI, agentic models and lab automation for scientific discovery. *Nature Reviews Methods Primers* (2025)
11. Papineni, K., Roukos, S., Ward, T., et al.: BLEU: a method for automatic evaluation of machine translation. In: *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*. pp. 311–318 (2002)
12. Richardson, D.S., Guo, Y., et al.: Unlocking the potential of large-scale 3D imaging with tissue optical clearing. *Journal of Microscopy* **74**(3), 179–192 (2025)
13. Singhal, K., Azizi, S., Tu, T., et al.: Towards expert-level medical question answering with large language models. *Nature* **620**, 172–179 (2023)
14. Ueda, H.R., Ertürk, A., Chung, K., et al.: Tissue optical clearing and its applications in neuroscience. *Nature Reviews Neuroscience* **21**(2), 61–79 (2020)
15. Wang, X., Li, M., Chen, W., et al.: CARDBiomedBench: a benchmark for evaluating large language models in biomedical applications. *The Lancet Digital Health* (2025)
16. Wang, Z., Li, M., Chen, X., et al.: Advances in tissue optical clearing for 3D imaging in large animals. *Photonics Insights* (2025)
17. Ying, J., Chen, Z., Wang, Z., Jiang, W., Wang, C., Yuan, Z., Su, H., Kong, H., Yang, F., Dong, N.: Seedbench: A multi-task benchmark for evaluating large language models in seed science. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 31395–31449 (2025)
18. Zhang, H., Chen, J., Wang, L., et al.: Benchmarking large language models for biomedical natural language processing. *Nature Communications* **16**, 56989 (2025)
19. Zhang, Y., Li, C., et al.: Survey and analysis of hallucinations in large language models. *Frontiers in Artificial Intelligence* (2025)